

PAPER_389 — Galactic ω_s Calibration from Stellar Velocity Dispersion

Source: grok_share_cfdcad2f5.txt, lines ~1-3200 (SMBH-UQFF comparison section)

Section: SMBH comparison to UQFF_17April2025.docx — galactic angular frequency derivation

Session: 106 (grok_share_cfdcad2f5.txt full analysis)

CP4 Class: GalacticOmegaSVelocityDispersionCalibrationCalculator (CP4 #40)

Abstract

This paper presents a UQFF analysis of Galactic ω_s Calibration from Stellar Velocity Dispersion, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

The UQFF framework encodes angular frequency inputs for astrophysical systems through parameters such as ω_s (system angular frequency) and ω_g (galactic angular velocity). These are central to the MUGE resonance channel and the buoyancy tier calculations.

The SMBH comparison to UQFF_17April2025.docx document introduces a direct observational calibration formula for the galactic angular frequency input $\omega_{s_galactic}$:

$$\omega_{s,galactic} = \frac{\sigma \times 10^3}{R_{bulge}}$$

Where: - σ = stellar velocity dispersion in km/s - $\times 10^3$ = unit conversion (km/s $\rightarrow$ m/s)
- R_{bulge} = bulge radius in meters

This formula bridges the directly observable M- σ relation parameters to the UQFF angular frequency input, making UQFF models directly anchored in spectroscopic observations.

2. Formula Derivation and Physical Basis

2.1 Dimensional Analysis

The angular frequency ω has SI units of rad/s. From a velocity dispersion:

$$[\omega] = \left[\frac{v}{R} \right] = \frac{\text{m/s}}{\text{m}} = \text{s}^{-1} \text{ (rad/s)}$$

Therefore $\omega_{s,\text{galactic}} = \sigma/R_{\text{bulge}}$ is dimensionally consistent when σ is in m/s and R_{bulge} is in m.

2.2 Physical Interpretation

- **σ (stellar velocity dispersion):** The 1D line-of-sight velocity dispersion of stars in a galactic bulge, measured from spectral line broadening. Typical values range from $\sigma \sim 100$ km/s (low-mass ellipticals) to $\sigma \sim 350$ km/s (massive BCGs).
- **R_{bulge} (bulge effective radius):** The half-light radius of the stellar bulge, from surface brightness photometry. Typical values: $R_{\text{bulge}} \sim 0.1\text{--}10$ kpc.
- **Physical meaning of $\omega_{s,\text{galactic}}$:** This angular frequency represents the characteristic orbital frequency of bulge stars (approximating circular orbit velocity as $v_{\text{circ}} \approx \sigma$). It sets the frequency scale for resonance terms in the MUGE 12-term resonance channel.

2.3 Relationship to Jeans/Virial Estimates

For a virially relaxed stellar system:

$$\sigma^2 \approx \frac{GM_{\text{bulge}}}{R_{\text{bulge}}}$$

Therefore:

$$\omega_{s,\text{galactic}} = \frac{\sigma}{R_{\text{bulge}}} \approx \frac{(GM_{\text{bulge}})^{1/2}}{R_{\text{bulge}}^{3/2}} = \sqrt{\frac{GM_{\text{bulge}}}{R_{\text{bulge}}^3}}$$

This is precisely Kepler's third law: $\omega_K = \sqrt{GM/R^3}$.

The formula $\omega_{s,\text{galactic}} = \sigma/R_{\text{bulge}}$ provides a **direct observational proxy for the Keplerian angular frequency** without requiring knowledge of the dynamical mass (which requires model-dependent assumptions), using only spectroscopic and photometric observables.

3. Worked Examples

3.1 Sagittarius A* (Milky Way Center)

From literature: - $\sigma = 100$ km/s (central stellar velocity dispersion) - $R_{\text{bulge}} = 1.5$ kpc
 $= 1.5 \times 3.086 \times 10^{19} \text{ m} = 4.629 \times 10^{19} \text{ m}$

$$\omega_{s,\text{galactic}}^{\text{SgrA}^*} = \frac{100 \times 10^3}{4.629 \times 10^{19}} = \frac{10^5}{4.629 \times 10^{19}} \approx 2.160 \times 10^{-15} \text{ rad/s}$$

This is consistent with the canonical $\omega_g = 7.3 \times 10^{-16}$ rad/s used in the UQFF pipeline (within a factor of ~ 3 , reflecting the difference between bulge rotation and nuclear disk).

3.2 M87 (Virgo A, NGC 4486)

From literature: - $\sigma = 324$ km/s (central 1'' aperture) - $R_{\text{bulge}} = 7$ kpc = 2.160×10^{20} m

$$\omega_{s,\text{galactic}}^{\text{M87}} = \frac{324 \times 10^3}{2.160 \times 10^{20}} = \frac{3.24 \times 10^5}{2.160 \times 10^{20}} \approx 1.5 \times 10^{-15} \text{ rad/s}$$

3.3 NGC 1275 (Perseus A)

From PAPER_259 context (BCG Perseus A): - $\sigma \approx 260$ km/s - $R_{\text{bulge}} \approx 5$ kpc = 1.543×10^{20} m

$$\omega_{s,\text{galactic}}^{\text{NGC1275}} = \frac{260 \times 10^3}{1.543 \times 10^{20}} \approx 1.685 \times 10^{-15} \text{ rad/s}$$

3.4 General Scaling

For a Milky-Way class spiral ($\sigma = 200$ km/s, $R_{\text{bulge}} = 2$ kpc):

$$\omega_{s,\text{galactic}}^{\text{MW-class}} = \frac{2 \times 10^5}{6.171 \times 10^{19}} \approx 3.24 \times 10^{-15} \text{ rad/s}$$

4. Calibration Table

System	σ (km/s)	R_{bulge} (kpc)	$\omega_{s,\text{galactic}}$ (rad/s)
Dwarf elliptical	40	0.3	4.24e-15
Milky Way center	100	1.5	2.16e-15
Milky Way spiral	200	2.0	3.24e-15
M87 (Virgo A)	324	7.0	1.50e-15
NGC 1275 (Perseus A)	260	5.0	1.69e-15
Massive BCG	350	15.0	7.56e-16

The canonical UQFF value $\omega_g = 7.3 \times 10^{-16}$ rad/s corresponds to a massive BCG ($\sigma \sim 350$ km/s, $R_{\text{bulge}} \sim 15$ kpc), consistent with its use in the CP1/CP3 outer-frame tidal calculation.

5. Integration into UQFF Pipeline

5.1 MUGE Resonance Term Update

The formula directly calibrates ω_s in the resonance MUGE:

```
def compute_omega_s_galactic(sigma_km_s: float, R_bulge_m: float) -> float:
    """
    Calibrate galactic angular frequency from observational parameters.

    Args:
        sigma_km_s: stellar velocity dispersion in km/s
        R_bulge_m: bulge radius in meters

    Returns:
        omega_s: angular frequency in rad/s
    """
    return (sigma_km_s * 1e3) / R_bulge_m
```

5.2 Replacing Hardcoded ω_g

For systems with known σ and R_{bulge} , this formula replaces the canonical $\omega_g = 7.3 \times 10^{-16}$ rad/s with a system-specific observationally anchored value.

The MUGE resonance tier term $\text{term_Ub_i} = -\beta_i \times \text{ug1} \times \omega_s \times (M/r) \times U_{\text{UA}} \times \cos(\pi \cdot t)$ then uses $\omega_{s,\text{galactic}}$ for the system-specific outer-frame coupling.

6. Observational Anchoring Chain

Complete chain from observation to UQFF:

Spectroscopic measurement: σ (SDSS/VLT/Keck line broadening)

↓

Photometric measurement: R_{bulge} (HST/ELT effective radius)

↓

Formula: $\omega_{s,\text{galactic}} = (\sigma \times 10^3) / R_{\text{bulge}}$

↓

MUGE resonance input: ω_s fed into 12-term co-sum

↓

PAPER_390 $M\text{-}\sigma$: $\log(M_{\text{BH}})$ calibrates SMBH mass input M

Together PAPER_389 (ω_s calibration) and PAPER_390 ($M_{\text{BH}}\text{-}\sigma$) provide a complete observational bridge for UQFF SMBH system parameterization.

7. Validation Cross-Reference

Reference	Connection
PAPER_390	M_BH- σ dispersion relation (companion observational anchor)
PAPER_339	Um rotor torque (uses ω in F_torque context)
PAPER_371	MUGE 12-term resonance (ω_s enters resonance co-sum)
PAPER_259	NGC1275 BCG (σ and R_bulge values cross-checked)
PAPER_346	M87 BZ-jet FUBi (σ and R_bulge values cross-checked)

Discovery Class: Observational calibration formula — first explicit $\sigma/R_{\text{bulge}} \rightarrow \omega_s$ mapping

Distinct from: PAPER_339 (torque context); all prior ω parameters (those are hardcoded constants, not observation-derived)

Key feature: Direct spectroscopic/photometric anchoring of UQFF angular frequency input; Kepler proxy